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CONVERTIBLE STEM/FRACTURE STEM

Abstract

A humeral anchor for a shoulder prosthesis is disclosed. The humeral anchor includes a distal portion configured to be anchored in a proximal region of a humerus and a proximal portion, the proximal portion including a proximal face, where the proximal face includes an engagement feature configured to directly couple to a reverse shoulder insert having a concave proximal portion, and wherein the proximal face is further configured to directly couple with an anatomical shoulder insert having a convex proximal portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of co-pending U.S. patent application Ser. No. 17/450,710, filed Oct. 13, 2021, which is a continuation of U.S. application Ser. No. 16/554,013, filed on Aug. 28, 2019, now U.S. Pat. No. 11,173,037, which is a division of U.S. application Ser. No. 15/532,035, filed on May 31, 2017, now U.S. Pat. No. 10,433,967, filed as application No. PCT/US2015/065126 on Dec. 10, 2015, the contents of which are incorporated by reference herein. Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 C.F.R. §§ 1.57; 1.97; and 1.98 in their entirety.

FIELD OF DISCLOSURE

[0002] The present application relates to apparatuses and methods for reverse and anatomic shoulder prostheses.

BACKGROUND

[0003] Arthroplasty is the standard of care for the treatment of shoulder joint arthritis. A typical anatomical shoulder joint replacement attempts to mimic anatomic conditions. For example, a metallic humeral stem and a humeral head replacement are attached to the humerus of the arm and replace the humeral side of the arthritic shoulder joint. Such humeral head replacement can articulate with the native glenoid socket or with an opposing glenoid resurfacing device.

[0004] For more severe cases of shoulder arthritis, the standard treatment is a reverse reconstruction, which includes reversing the kinematics of the shoulder joint. A reverse shoulder prosthesis can be provided by securing a semi-spherical device (sometimes called a glenoid sphere) to the glenoid and implanting a humeral stem with a cavity capable of receiving the glenoid sphere.

[0005] As patient disease may progress after anatomic treatment, revision surgery may be necessary to perform a reverse reconstruction of the shoulder. In the known art, the change in the type of prosthesis is addressed either below the plane of resection or above the plane of resection. In prosthesis that are converted from anatomic to reverse by a modularity below the plane of resection, removal of anatomic devices that have integrated into the patient's bony anatomy proves to be difficult for the surgeon, and could potentially cause excessive patient bone loss. One advantage of such conversion is that the reverse insert could partially reside below the resection plane and therefore reduce the distance between the cavity and the lateral contour of the humerus. Such position has proven to be beneficial to a reversed kinematics. In contrary, in prosthesis that are converted from anatomic to reversed above the plane of resection thanks to an adaptor, reverse kinematic is altered as the position of the cavity is further push out of the humerus by the addition

of the adaptor above the resection plane. Such constructs are typically made of 3 components that present an extra modularity in comparison to 2 components construct and could potentially cause disassembly or breakage of the construct. One possibility to limit the alteration of the kinematics and limit the modularity is to inverse the bearing surface material by having a harder cavity within the humerus and a softer semi-spherical device secured to the glenoid. But the proven clinical design and preferred embodiment is usually that the cavity is softer than the semi-spherical device. [0006] In cases of displaced or dislocated 3- and 4-part proximal humeral fractures, the proximal humerus also needs to be reconstructed. Although hemi-arthroplasty procedures may be used for the treatment of such displaced fractures, the functional outcomes of these procedures are often reported as poor and unpredictable.

SUMMARY

[0007] A convertible prosthesis that can be converted from an anatomic replacement to a reverse reconstruction without removal of parts integrated into the patient's bony anatomy is highly desirable. For improved patient outcomes, such a convertible prosthesis should respect the biomechanics of a true anatomic replacement while also performing well when converted into a reverse reconstruction. In some cases, it may also be desirable for the convertible prosthesis to be configured for use in a humeral fracture repair procedure.

[0008] The aim of the present invention is to provide efficient surgical means that are appropriate to this problematics.

[0009] To that end, the invention relates to a kit for a shoulder prosthesis, comprising: [0010] a humeral anchor comprising a proximal portion and a distal portion, the proximal portion including a proximal face, the proximal face comprising a hole and a cavity that is distinct from the hole; [0011] a reverse insert having a proximal portion and a distal portion, the proximal portion including a concave surface configured to receive a glenosphere and the distal portion comprising a protrusion, wherein the reverse insert is configured to directly couple to the cavity of the proximal face; and [0012] an anatomical insert having a proximal portion including a convex surface and a distal portion including a protrusion, wherein the anatomical insert is configured to directly couple to the hole of the proximal face.

[0013] More generally, according to some embodiments of the present disclosure, a convertible prosthesis system includes a humeral anchor, an anatomic or humeral head insert, and a reverse insert. The system advantageously has a 2-part construction for each of the anatomic configuration and the reverse configuration, with one of the parts common to both configurations. The modularity of the system is advantageously located at a humeral resection plane. In some embodiments, the humeral anchor is a fracture stem configured for use in a humeral fracture repair procedure.

[0014] In some embodiments, the anatomical insert and the reverse insert are made out of different materials but each being monolithic. In some embodiments, the reverse insert resides partially below the resection plane.

[0015] According to additional advantageous features of the set according to the invention, considered alone or according to all technically possible combinations: [0016] the anatomical insert is configured to rotationally engage the proximal face; [0017] the protrusion of the anatomical insert is configured to rotationally engage the hole of the proximal face; [0018] the humeral anchor is a fracture stem; [0019] the humeral anchor comprises a stem comprising a metaphyseal portion comprising a medial arm and first and second lateral arms extending between and connecting the distal portion and the proximal portion of the stem; [0020] the kit further comprises a bone graft; [0021] the bone graft is shaped to be received thereon the stem; [0022] the bone graft is selected from the group consisting of bone, stem cells, ceramic, polymer and porous metal; [0023] the bone graft is selected from the group consisting of an allograft and autograft; [0024] the proximal portion of the humeral anchor comprises a spherical portion; [0025] the proximal face further comprises a groove extending around an inner periphery of the cavity; [0026] the protrusion of the reverse insert further comprises a locking member configured to engage the groove of the proximal

face; [0027] the proximal face further comprises a ridge defining an inner periphery of the cavity smaller than an outer periphery of the protrusion of the distal portion of the reverse insert, whereby an interference fit is provided between the protrusion of the distal portion of the reverse insert and the cavity when the reverse insert is engaged with the humeral anchor; [0028] an interference fit is provided between the protrusion of the distal portion of the reverse insert and the cavity when the reverse insert is engaged with the humeral anchor; [0029] a periphery of the cavity is spaced from and surrounds the hole configured to receive the protrusion of the anatomical insert; [0030] the hole configured to receive the protrusion of the anatomical insert is at least partially formed in a raised portion of the proximal face, the raised portion extending proximally from a base of the cavity; [0031] a distal end of the protrusion of the reverse insert comprises a recess configured to engage the raised portion of the stem face; [0032] an interface between the raised portion of the proximal face and the recess of the reverse insert is configured to resist rotation between the reverse insert and the humeral anchor; [0033] a central axis extending proximally and distally through the hole is offset from a central axis extending proximally and distally through the cavity; [0034] the reverse insert is configured to directly couple to the proximal face via a snap-fit; [0035] the humeral anchor comprises a unitary body; [0036] the distal portion of the humeral anchor comprises a taper; [0037] the humeral anchor comprises a base member comprising a distal end configured to be embedded in bone and a proximal end to be disposed at a bone surface, the base member having a plurality of spaced apart arms and a concave member comprising the hole projecting from the proximal end toward the distal end; and an anchor component comprising the proximal face and having a distal portion advanceable into the base member to a position disposed within the arms, the distal portion of the anchor component configured to project circumferentially into a space between the arms, the distal portion of the anchor component being exposed between the arms when the anchor component is advanced into the base member; wherein the cavity is defined at least in part by the proximal face of the anchor component; [0038] the proximal face of the anchor component comprises an aperture configured to be advanced over a proximal portion of the concave member; [0039] the proximal face comprises a driver interface disposed thereon outward of the aperture; [0040] the distal portion of the anchor component comprises a cylindrical sleeve and a thread projecting laterally therefrom; [0041] the kit further comprises a locking device disposed between the anchor component and the base member to prevent disengagement of the anchor component from the base member.

[0042] The invention further relates to a method of using the kit as defined above, this method comprising selecting intra-operatively to implant the reverse shoulder insert or the anatomical shoulder insert.

[0043] The invention further relates to a stem for a shoulder prosthesis is provided that includes a distal shaft portion, a proximal portion, and a metaphyseal portion. The distal shaft portion is adapted to be anchored in a medullary canal of a humerus. The proximal portion includes a stem face. The stem face includes a first engagement feature configured to directly couple with a reverse insert and a second engagement feature configured to couple with an anatomical insert. Further, the first engagement feature is distinct from the second engagement feature. The metaphyseal portion includes a medial portion and first and second lateral arms extending between and connecting the shaft portion and the proximal portion.

[0044] In some variations of the stem, the second engagement feature is configured to directly couple with the anatomical insert. In some embodiments, the stem further includes a notch configured to engage a suture. The stem can further include a fin protruding from a lateral side of the shaft portion and extending from a proximal portion of the shaft portion distally along a portion of a length of the shaft portion. In some embodiments, the medial portion includes an arm having a lateral edge, the first and second lateral arms have medial edges, and a fenestration is defined between the lateral edge of the medial arm and the medial edges of the first and second lateral arms. In some embodiments, the first lateral arm has a first inner edge, the second lateral arm has a

second inner edge, the first and second inner edges face one another, and a gap is formed between the first and second inner edges. In some such embodiments, the gap extends from a proximal end of the shaft portion to a distal end of the proximal portion. In that case, the stem may further comprises a bone graft shaped to be received thereon the gap from lateral wherein the graft fill the fenestration and contact the lateral edge of the medial arm. If any, the graft extends laterally above the lateral edges of the first and second lateral arms. In variation, the graft extends laterally from the gap to cover the lateral edge and the outer edge, opposite to the inner edge, of the first lateral arm and to cover the lateral edge and the outer edge, opposite to the inner edge, of the second lateral arm.

[0045] The invention further relates to a humeral anchor for a shoulder prosthesis is provided. The humeral anchor has a distal portion and a proximal portion. The distal portion is configured to be anchored in a proximal region of a humerus. The proximal portion includes a proximal face. The proximal face includes an engagement feature configured to directly couple to a reverse should insert having a concave proximal portion. The proximal face is also configured to directly couple with an anatomical shoulder insert having a convex proximal portion.

[0046] In some variations, a kit including the humeral anchor further includes a reverse insert that has a proximal portion that includes a concave surface configured to engage a glenosphere and a distal portion that is configured to directly attach to the engagement feature of the proximal face. In some embodiments, the proximal face further includes a cavity having an inner dimension, the distal portion of the reverse insert has an outer dimension, and an interference fit is provided between the distal portion of the reverse insert and the cavity when the reverse insert is engaged with the humeral anchor. In some embodiments, the kit further includes an anatomical insert that has a proximal portion including a convex surface and a distal portion that is configured to couple to the proximal face.

[0047] In some embodiments, the humeral anchor includes: a base member comprising a distal end configured to be embedded in bone and a proximal end to be disposed at a bone surface, the base member having a plurality of spaced apart arms and an anchor component having a proximal end and a distal portion advanceable into the base member to a position disposed within the arms, and the distal portion of the anchor component comprises threads configured to project circumferentially into a space between the arms, the threads being exposed between the arms when the anchor component is advanced into the base member. In some embodiments, the humeral anchor further includes a concave member comprising a hole configured to receive a distal shaft of an anatomical shoulder insert. In some embodiments, the anchor component at least partially defines a cavity configured to engage a reverse shoulder insert.

[0048] The invention further relates to a method for shoulder surgery is provided. In this method, a humeral anchor is provided. The humeral anchor includes a distal portion and a proximal portion. The proximal portion includes a proximal face that is configured to directly couple to a reverse shoulder insert and an anatomical shoulder insert. A surgeon or other user chooses intra-operatively to implant the reverse shoulder insert or the anatomical shoulder insert. The surgeon or other user implants a two-component shoulder system that is selected from the group consisting of a humeral anchor directly attached to the reverse shoulder insert and a humeral anchor directly attached to the anatomical shoulder insert.

[0049] The invention further relates to a method for shoulder surgery is provided. In this method, a humeral anchor of a humeral component shoulder system is implanted at least partially in a proximal portion of a humerus. The humeral anchor is adapted to directly interface with a one-component reverse shoulder insert and an anatomical shoulder insert. A surgeon or other user chooses intra-operatively to implant a one-component reverse shoulder insert or an anatomical shoulder insert. The surgeon or other user directly couples the chosen insert to the humeral anchor.

[0050] The invention also relates to a method for shoulder surgery is provided. In this method, a humeral anchor is disposed at least partially in a proximal portion of a humerus. A surgeon or other

user selects a reverse shoulder insert or an anatomical shoulder insert. The reverse shoulder insert includes a body with a concave articular surface on one side and an engagement structure projecting from a side of the body opposite the concave surface. The surgeon or other user implants the insert that has been chosen by directly coupling the chosen insert to the humeral anchor.

[0051] The invention further relates to a method for revision shoulder surgery is provided. In this method, an anatomical shoulder insert is removed from a humeral anchor implanted in a patient's humerus to expose a proximal face of the humeral anchor. A reverse shoulder insert is directly coupled to the proximal face.

[0052] The invention further relates to a kit for a shoulder prosthesis that includes a stem and a bone graft cutter. The stem includes a unitary body having a shaft portion configured to be anchored in a medullary canal of a humerus and a proximal portion. The proximal portion includes a stem face that includes an engagement feature configured to directly couple to a reverse shoulder insert having a concave proximal portion. The stem face is further configured to directly couple with an anatomical shoulder insert having a convex proximal portion.

[0053] In some variations, the bone graft cutter includes a drill guide and a cutting cap. In some such variations, the drill guide includes at least 4 holes provided at different angles. The cutting cap can be shaped to be received therein the drill guide. In some variations, the kit includes a drill, mallet, and/or bone graft. The bone graft can be shaped to be received thereon the stem. The bone graft can be selected from the group consisting of bone, stem cells, ceramic, polymer and porous metal. The bone graft can be selected from the group consisting of an allograft and autograft.

[0054] The invention further relates to a kit for a shoulder prosthesis is provided that includes a humeral anchor and an anchor holder. The humeral anchor includes a distal portion configured to be anchored in a proximal portion of a humerus and a proximal portion. The proximal portion includes a proximal face that includes an engagement feature configured to directly couple to a reverse shoulder insert having a concave proximal portion. The proximal face is further configured to directly couple with an anatomical shoulder insert having a convex proximal portion.

[0055] In some variations of this kit, the anchor holder is monolithic. The anchor holder can be deformable. In some variations, the anchor holder includes at least one compression feature configured to retain the engagement feature of the proximal face. The anchor holder can be in a snap-fit arrangement with the face. The anchor holder can be in an interference fit arrangement with the face.

[0056] In some embodiments, a kit for a shoulder prosthesis comprises: [0057] a humeral anchor comprising a proximal portion and a distal portion, the proximal portion including a proximal face; [0058] a reverse insert having a proximal portion and a distal portion, the proximal portion including a concave surface configured to receive a glenosphere and the distal portion comprising a protrusion; and [0059] an anatomical insert having a proximal portion including a convex surface and a distal portion including a protrusion; [0060] characterized in that the proximal face comprises a hole and a cavity that is distinct from the hole, the reverse insert is configured to directly couple to the cavity of the proximal face, and the anatomical insert is configured to directly couple to the hole of the proximal face.

[0061] Any feature, structure, or step disclosed herein can be replaced with or combined with any other feature, structure, or step disclosed herein, or omitted. Further, for purposes of summarizing the disclosure, certain aspects, advantages, and features of the inventions have been described herein. It is to be understood that not necessarily any or all such advantages are achieved in accordance with any particular embodiment of the inventions disclosed herein. No aspects of this disclosure are essential or indispensable.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0062] These and other features, aspects and advantages are described below with reference to the drawings, which are intended to illustrate but not to limit the inventions. In the drawings, like reference characters denote corresponding features consistently throughout similar embodiments. The following is a brief description of each of the drawings.

[0063] FIG. 1A is a side plan view of a reverse shoulder prosthesis;

[0064] FIG. 1B is a side plan view of another embodiment of a reverse shoulder prosthesis;

[0065] FIG. 2A is a side plan view of an anatomic shoulder prosthesis;

[0066] FIG. 2B is a side plan view of another embodiment of an anatomic shoulder prosthesis;

[0067] FIG. 3 is a perspective view of a humeral stem that is used in the shoulder prostheses of FIGS. 1A and 2A;

[0068] FIG. 4A is a first view of the stem of FIG. 3;

[0069] FIG. 4B is a second side view of the stem of FIG. 3;

[0070] FIG. 4C is a third, alternate view of the stem of FIG. 3;

[0071] FIG. 4D is a cross-sectional view of the stem of FIG. 3 taken through the section plane 4D-4D in FIG. 4B;

[0072] FIG. 5 is a cross-sectional view of the stem of FIG. 3 taken through the section plane 5-5 in FIG. 4A;

[0073] FIG. 5A is a detail cross-sectional view of a portion of another embodiment of a stem;

[0074] FIG. 6 is a plan view of one embodiment of a face of the stem of FIG. 3, the face being configured to couple with a reverse insert to form the reverse shoulder prosthesis of FIG. 1A and with an anatomic insert to form the anatomic shoulder prosthesis of FIG. 2A.

[0075] FIG. 7 is an exploded view of the reverse shoulder prosthesis of FIG. 1A;

[0076] FIG. 8 is a cross-sectional view of the reverse shoulder prosthesis of FIG. 1A taken through section plane 8-8 shown in FIG. 1A;

[0077] FIG. 8A is a detail cross-sectional view of another embodiment of a reverse shoulder prosthesis including the stem of FIG. 5A;

[0078] FIG. 8B is a detail view of a portion of the reverse shoulder prosthesis of FIG. 8A showing an interference fit between the stem and the reverse insert.

[0079] FIG. 9 is a bottom perspective view of the reverse insert of the reverse shoulder prosthesis of FIGS. 1A and 1B;

[0080] FIG. 10A is a bottom view of the reverse insert of FIG. 9;

[0081] FIG. 10B is a top view of the reverse insert of FIG. 9;

[0082] FIG. 11 is a first side view of the of the reverse insert of FIG. 9;

[0083] FIG. 12 is cross-sectional view of the reverse insert of FIGS. 9-11 taken through the section plane 12-12 in FIG. 11;

[0084] FIG. 13 is an exploded view of the anatomic shoulder prosthesis of FIGS. 2A and 2B;

[0085] FIG. 14 is a cross-sectional view of the anatomic shoulder prosthesis of FIG. 2A taken through section plane 14-14 shown in FIG. 2A;

[0086] FIG. 15 is a side plan view of the anatomic insert of the anatomic shoulder prosthesis of FIGS. 2A and 2B;

[0087] FIG. 16 is a bottom perspective view of the anatomic insert of FIG. 15;

[0088] FIG. 17A is bottom view of the anatomic insert of FIG. 15;

[0089] FIG. 17B is a top view of the anatomic insert of FIG. 15;

[0090] FIG. 18 is a side view of a humerus;

[0091] FIGS. 19-22 illustrate the use of the reverse and anatomic shoulder prostheses of FIGS. 1A and 2A, for example, in an anatomic-to-reverse conversion procedure;

[0092] FIG. 23 illustrates a stemless anatomic shoulder prosthesis;

[0093] FIG. 23A is a top perspective view of a stemless humeral anchor;

[0094] FIG. 23B is a cross-sectional view of the stemless humeral anchor of FIG. 23A, taken at the section plane 23B-23B;

[0095] FIG. 23C is an exploded view of components of certain embodiments of the stemless humeral anchor of FIG. 23A shown in the context of the proximal humerus;

[0096] FIG. 23D illustrates the performance of various embodiments of stemless humeral shoulder assemblies similar to those illustrated in FIGS. 23-23C;

[0097] FIG. 24 illustrates a perspective view of a stem holder;

[0098] FIG. 25A illustrates a perspective view of a different embodiment of a stem holder;

[0099] FIG. 25B illustrates a perspective view of the stem holder of FIG. 25A engaging a stem of the present disclosure;

[0100] FIGS. 26A-E illustrate a perspective views of a bone cutter of the present disclosure;

[0101] FIGS. 27A-C illustrate a perspective views of a bone graft of the present disclosure; and

[0102] FIGS. 28A-B illustrate a perspective view of a fracture stem receiving the bone graft of FIG. 27C.

DETAILED DESCRIPTION

[0103] While the present description sets forth specific details of various embodiments, it will be appreciated that the description is illustrative only and should not be construed in any way as limiting. Furthermore, various applications of such embodiments and modifications thereto, which may occur to those who are skilled in the art, are also encompassed by the general concepts described herein. Each and every feature described herein, and each and every combination of two or more of such features, is included within the scope of the present invention provided that the features included in such a combination are not mutually inconsistent.

[0104] FIG. 1A illustrates a modular reverse shoulder prosthesis **10**, and FIG. 2A illustrates a modular anatomic shoulder prosthesis **20**, according to embodiments of the present disclosure. The reverse shoulder prosthesis **10** includes a stem **30** and a reverse insert **12**. The stem **30** is one embodiment of a humeral anchor disclosed herein. In embodiments, the anatomic shoulder prosthesis **20** includes the same stem **30** and an anatomic insert or humeral head **22**. FIG. 1B illustrates another example embodiment of a modular reverse shoulder prosthesis **10'** including a stem **30'** and reverse insert **12**, and FIG. 2B illustrates another example embodiment of a modular anatomic shoulder prosthesis **20'** including the same stem **30'** and the anatomic insert or humeral head **22**. The stem **30'** is another embodiment of a humeral anchor disclosed herein. The stem **30**, **30'**, anatomic insert **22**, and reverse insert **12** can be made of various materials, for example, metal, such as titanium, a ceramic material, ultra-high molecular weight polyethylene (UHMWPE), or other materials, and may include bioactive coatings such as HA or vitamin E, each of which may be porous or non-porous. In one embodiment the reverse insert is monolithic and comprise one unique material softer than the one comprised in the anatomic insert. More specifically, the reverse insert may comprise UHMWPE.

[0105] FIGS. 3-6 illustrate one embodiment of the stem **30** in greater detail. The stem **30** of FIGS. 1A and 2A is a fracture stem configured to be used in humeral fracture repair procedures as described herein. The stem **30'** of FIGS. 1B and 2B is a non-fracture stem and therefore may not include all of the features of stem **30** described herein, but may include other features of the stem **30**. The stem **30** is configured to be anchored in a medullary canal of a humerus of a patient. The stem **30** includes a shaft or distal portion **32** and a proximal portion **34**. In some embodiments, the stem **30** is a unitary body. In some embodiments, a humeral anchor is provided that has a unitary body. In other words, the stem **30** is monolithic, and the distal portion **32** and proximal portion **34** are integrally formed. In some embodiments, the stem **30**, the stem **30'**, and/or other humeral anchors herein can have a distal portion that includes a taper. For example, the shaft portion **32** can have a gradually tapered overall shape to better fit the humerus bone into which it is implanted. A length of the distal portion **32** of the stem **30**, **30'** can vary. In some embodiments, a modular reverse shoulder prosthesis and a modular anatomic shoulder prosthesis including various features

as described herein can be stemless. In other words, the prosthesis need not include the shaft or distal portion **32**. For example, FIG. **23** illustrates an example embodiment of a stemless anatomic shoulder prosthesis including a humeral anchor **30''** and the anatomic insert **22**. The humeral anchor **30''** is another embodiment of a humeral anchor disclosed herein.

[0106] In some embodiments, the proximal portion **34** includes a spherical portion. For example, in the illustrated embodiment, the outer surface **35** of the proximal portion **34** is shaped generally as a half-sphere. A proximal end of the proximal portion **34** includes a stem face **36**. The stem face **36** is one example of a proximal face as disclosed herein. The stem **30'** may also have the features of the stem face **36**. The humeral anchor **30''** has a proximal portion that may include some or all of the features of the stem face **36**. For example, the stem face **36** and the proximal face of the humeral anchor **30''** can each have a hole for mounting an anatomic insert and a cavity for mounting a reverse insert, as discussed in greater detail below. The stem **30'** of the reverse shoulder prosthesis **10'** of FIG. **1B** and the anatomic shoulder prosthesis **20'** of FIG. **2B** can include any or all of the features of the stem face **36** of stem **30** described and shown herein. A proximal end of the humeral anchor **30''** of the stemless design shown in FIG. **23** can also include any or all of the features of the stem face **36** described and shown herein. A distal portion of the humeral anchor **30''** can include a taper as discussed further below.

[0107] The stem face **36** is oriented at an angle relative to a longitudinal stem axis **31**, shown in FIG. **4B**. An obtuse angle between an axis A (also shown in FIG. **4B**) normal to the plane of the stem face **36** and the stem axis **31** can be in the range of about 120° to about 150°. For example, in the illustrated embodiment, the angle between the normal axis A and the stem axis **31** is about 132.5°. The angle can be selected to allow the stem **30** to provide desirable biomechanics when used in the anatomic shoulder prosthesis **20** as well as to perform well when used in the reverse shoulder prosthesis **10**. Other angles are also possible, for example, 127.5°, 137.5°, and other angles. The reverse insert **12** can also be made with various inclination angles as described in greater detail herein to produce a reverse shoulder prosthesis **10** having a variety of overall angles. For example, the overall angle of the prosthesis **10** can be in the range of 130° to 155°. The angle of the stem face **36** can be selected based on pre-operative imaging of the patient's humeral bone. The inclination angle and thickness of the reverse insert **12** can be selected based on a numerical simulation of range of motion using virtual surgery. The inclination angle can be selected to achieve an optimized range of motion, which results in no or minimized contact between a medial side of the insert **12** and the scapula pillar in any movement (e.g., adduction, flexion-extension, internal and external rotation) and no or minimized limitation due to contact between the humerus and scapula during arm elevation.

[0108] Whereas some available shoulder prosthesis systems require an adapter or spacer to couple either or both of a reverse insert and an anatomic insert to a stem, the stem face **36** of the present disclosure advantageously is configured to couple or attach directly to either of the reverse insert **12** and the anatomic insert **22** without the need for a spacer, adapter, or the like. The stem face **36** includes a first engagement feature configured to directly couple with the reverse insert **12** and a second engagement feature configured to couple with the anatomic insert **22**. In the illustrated embodiment, the first engagement feature includes a cavity **40** recessed from a peripheral rim **38** of the stem face **36**, and the second engagement feature includes a hole or female tapered recess **42**. The first engagement feature is distinct from the second engagement feature. More specifically, the cavity **40** is distinct from the hole or the female tapered recess **42**. Having two engagement features offer several advantages. The first advantage is to adapt each engagement feature to the specific properties of the most suitable material for the reverse insert and to the specific properties of the most suitable material for the anatomic insert. In contrary, having only one engagement features only obliged to alter the material of one of the insert or to alter the engagement feature of one of the insert. Both of those alteration might cause unsatisfactory issues such as increased wear, disassembly, breakage and the like. The second advantage is to adapt the position of each

engagement features to the most suitable position. For example, it is known in the prior art that the most suitable position for the cavity of the reverse insert is to partially reside below the resection plane while the most suitable position for the convexity of the anatomic insert is to reside above the resection plane.

[0109] A periphery or sidewall **41** of the cavity **40** is radially spaced from and surrounds the hole **42**. The outer periphery of the cavity **40** surrounded by the sidewall **41** can be circular or generally circular as shown, although other shapes and configurations are also possible. The stem face **36** can further include a groove **46** extending around an inner periphery or circumference of the cavity **40**. As shown, the groove **46** is recessed in the sidewall **41** of the cavity **40** and therefore extends radially outwardly from the cavity **40**. The hole **42** can be at least partially formed in a raised portion **44** of the stem face **36**. The raised portion **44** extends proximally from a base **43** of the cavity **40**. As shown, a proximal surface **45** of the raised portion **44** can be in the same plane or substantially the same plane as the peripheral rim **38**. In the illustrated embodiment, the outer perimeter of the raised portion **44** has a non-circular and radially asymmetric shape. As shown in FIG. 6, in various embodiments the raised portion **44** has at least one flat peripheral edge. For example, the raised portion **44** can have a curved base **142**, one or more angled sides **144** that angle outwardly from the curved base **142**, and a straight top surface **146**. Alternatively, in some embodiments where rotation of the reverse insert is desired, the raised portion **44** can have a shape that is circular or non-circular but rotationally symmetric. In some embodiments, the sidewall **41** of the cavity **40** can be non-circular but rotationally symmetric. For example, the sidewall **41** can have a rotationally symmetric cross-sectional shape similar to that of the metaphysis described in PCT Publication No. WO 2013/064569, the entirety of which is hereby incorporated by reference herein.

[0110] FIG. 5A illustrates an alternative embodiment of a stem face **36'**. The stem face **36'** is similar to the stem face **36** except as described differently below. The stem face **36'** can be incorporated into any humeral anchor according to the present disclosure, including the stem **30**, the stem **30'**, the humeral anchor **30''**, or the humeral anchor **200**. As shown, the stem face **36'** includes one or more ridges **47** extending radially inwardly into the cavity **40** from the sidewall **41**. In the illustrated embodiment, the ridges **47** are integrally formed with the sidewall **41**. In other words, the sidewall **41** and ridges **47** are monolithic. However, in other embodiments, the ridges **47** can be formed separately from and coupled, removably or permanently, to the sidewall **41**. In the illustrated embodiment, the stem face **36'** includes three ridges **47**, although more or fewer ridges **47** are also possible. The ridges **47** (e.g., the peaks **47A** or innermost points of the ridges **47**) define an inner dimension of the cavity **40**. In the illustrated embodiment, the ridges **47** are generally triangular. As shown, a distal surface of the ridge(s) **47** can extend perpendicular or generally perpendicular to the sidewall **41** and/or the base **43** of the cavity **40**, and a proximal surface of the ridge(s) **47** can extend at an angle relative to the sidewall **41**, base **43**, and/or distal surface of the ridge(s) **47**. A junction between the distal surface of a first ridge **47** and the proximal surface of a second ridge **47** that is adjacent and distal to the first ridge **47** can be rounded as shown, or may be more angular to form a relatively sharp corner. Other shapes and configurations for the ridges **47** are also possible. In some embodiments, the ridges **47** extend around the entire sidewall **41**. In other embodiments, the ridges **47** can extend only partially around the sidewall **41**.

[0111] FIGS. 9-12 illustrate the reverse insert **12** in greater detail. The reverse insert **12** is configured to couple with the stem face **36** by virtue of features shown in the exploded view of FIG. 7, assembled view of FIG. 1A, and assembled section view of FIG. 8. The reverse insert **12** includes a distal portion **50** and a proximal portion **52**. The proximal portion **52** can include a concave proximal surface **54** configured to interface with a glenosphere or the like, which can be implanted in a patient's glenoid as part of a reverse prosthesis surgery. An outer wall **56** of the reverse insert **12** can be substantially cylindrical.

[0112] As shown in FIGS. 11 and 12, a top or proximal surface of the reverse insert **12** is angled relative to a bottom or distal surface. The angle between the bottom surface and the top surface is

the inclination angle of the insert. The reverse insert **12** can be provided in a number of different inclination angles, for example 7.5°, 12.5°, and 17.5°. Other inclination angles are also possible. In some embodiments, the user can select among stems **30** having various angles between the normal axis A and the stem axis **31** as described herein and among reverse inserts **12** having various inclination angles to create a reverse shoulder prosthesis **10** having an overall angle selected and suited for the particular patient. For example, a reverse insert **12** having an inclination angle of 12.5° can be used with a stem **30** having an angle of 132.5° to create an overall angle of 145°.

[0113] The distal portion **50** of the reverse insert **12** includes a protrusion **58**. The protrusion **58** can be integrally formed with the proximal portion **52** of the reverse insert **12**. In other words, the proximal portion **52** and protrusion **58** are monolithic or unitary. The protrusion **58** is configured to directly interface and couple with the cavity **40** of the stem face **36**. For example, a peripheral surface of the protrusion **58** can form one part of a direct interface between the insert **12** and the stem face **36**. Another part of the direct interface can include all or a portion of the sidewall **41** of the cavity **40**. By providing the direct interface between the insert **12** and the stem **30**, once the distal portion **32** of the stem **30** is lodged in the humerus, the insert **12** can be immediately applied to the stem **30** without the need for assembling a metaphysis or other intervening component. In some embodiments, the protrusion **58** couples with the cavity **40** via a snap-fit or friction fit. In the illustrated embodiment, the protrusion **58** is circular and configured to couple with the circular cavity **40** shown in FIGS. 3 and 6. Alternatively, in other embodiments, the protrusion **58** can be non-circular but rotationally symmetric to couple with a non-circular but rotationally symmetric sidewall **41**. For example, the protrusion **58** can be shaped similarly to the locking protrusion of the reverse insert described in PCT Publication No. WO 2013/064569. As shown, the protrusion **58** can include an indexing marker **64** configured to assist with properly aligning the reverse insert **12** with the stem face **36**.

[0114] The protrusion **58** can include one or more locking members. In the illustrated embodiment, the protrusion **58** includes a C-ring **60** for mechanically coupling the reverse insert **12** directly to the stem **30**. As shown in FIG. 7, the C-ring **60** is formed separately from the reverse insert **12** and is snapped, pressed, or otherwise placed onto the protrusion **58** either permanently or removably. In other embodiments, the C-ring **60** or another locking member can be integrally formed with the protrusion **58**. In the illustrated embodiment, the C-ring **60** includes a gap **62** that enables the C-ring to flex to facilitate insertion of the protrusion **58** into the cavity **40**. The gap **62** can be aligned with the indexing marker **64**. In some embodiments, the locking member is an O-ring or protrusion that extends around an entire outer periphery of the protrusion **58**. In some embodiments, the locking member or members can be other deflectable members that can project away from the protrusion **58** less in one state to permit the protrusion **58** to be advanced into the cavity and can project away from the protrusion **58** more in another state to enable the locking member or members to project into engagement with the sidewall **41**, e.g., into the groove **46**. The C-ring **60** or other locking member is configured to engage or couple with the groove **46** of the stem face **36**. The C-ring **60** can include a gradually ramped distal portion **68**, and a perpendicular or right-angle stop portion **70** at a proximal end. The ramped portion **68** has a smaller diameter toward a distal portion thereof and a progressively larger diameter toward a proximal portion thereof. The ramped portion **68** allows the C-ring **60** to slide into the groove **46**, and the stop portion **70** inhibits disengagement or removal of the C-ring **60** from the groove **46**. In some embodiments, the C-ring **60** allows the reverse insert **12** to couple with the stem face **36** via a snap-fit.

[0115] In the embodiment of the stem face **36'** shown in FIG. 5A, the ridge(s) **47** can act as locking member(s) or feature(s). The inner periphery of the cavity **40** of the stem face **36'** defined by the ridges **47** can be smaller than the outer periphery or an outer dimension of the protrusion **58**. An interference fit can therefore be provided between the protrusion **58** and the cavity **40**, e.g., the ridges **47**, when the reverse insert **12** is engaged with or coupled to the stem **30** (as shown in FIGS. 8A-8B) or **30'** or the humeral anchor **30''** or **200**. In particular, the outer periphery of the protrusion

58 of the reverse insert **12** is larger than a dimension of the cavity **40** defined transversely across the cavity **40** between peaks **47A** of the ridges **47** on opposite sides of the cavity **40**. The peaks **47A**, where provided can be pointed or rounded in different embodiments. The outer periphery of the protrusion **58** of the reverse insert **12** is smaller than the dimension defined by the base of the ridge **47** or defined by the sidewall **41** from which the ridges **47** extend. As a result, the insert **12** can be inserted into the cavity **40** but with some interference with the peaks **47A** of the ridges **47**, providing an interference fit. In some embodiments, ridges could alternatively or additionally be provided on the protrusion **58** of the insert **12**. In such an embodiment, the protrusion **58** may not include a C-ring **60** or other locking member, as shown in FIGS. **8A-8B**. In some embodiments, the stem face **36**, **36'** and/or protrusion **58** can include other members or features that provide an interference fit between the protrusion **58** and the cavity **40** when the reverse insert **12** is engaged with or coupled to a humoral anchor.

[0116] A distal end of the protrusion **58** can include a recess **72**. The recess **72** is configured to engage or receive the raised portion **44** of the stem face **36**, for example, as shown in FIG. **8**. In the illustrated embodiment, the recess **72** has a non-circular and radially asymmetric shape configured to correspond to the non-circular and radially asymmetric shape of the raised portion **44**. As shown in FIG. **10A**, the recess **72** can have a curved side **152** configured to correspond to the curved base **142** of the raised portion **44**, angled edges **154** that angle outwardly from the curved side **152** and are configured to correspond to the angled sides **144** of the raised portion **44**, and a straight side **156** configured to correspond to the straight top **146** of the raised portion. The shape of the recess **72** and raised portion **44** can help inhibit or resist rotation between the reverse insert **12** and the stem **30**. For example, one or both of the recess **72** and the raised portion **44** can have at least one non-circular portion, e.g., a straight portion of a periphery to inhibit rotation of the insert **12** relative to the stem **30**. The recess **72** and the raised portion **44** can have mating straight edge portions to inhibit such rotation. The recess **72** and the raised portion **44** can have multiple pairs of mating or matched straight edge portions to inhibit such rotation. The radially asymmetric shapes or rotation inhibiting configurations can help guide the surgeon in properly aligning or orienting the reverse insert **12** relative to the stem **30**. Examples of alternative configurations for resisting rotation between the reverse insert **12** and the stem **30** are described in WO 2014/067961, the entirety of which is hereby incorporated by reference herein. Alternatively, in other embodiments, the recess **72** and raised portion **44** can have corresponding non-circular but rotationally symmetric shapes.

[0117] FIGS. **15-17B** illustrate the anatomic insert **22** in greater detail. The anatomic insert **22** is configured to directly couple with the stem face **36** as shown in the assembled view of FIG. **2A** and assembled section view of FIG. **14** by advancing the anatomic insert **22** along axis B shown in the exploded view of FIG. **13**. The anatomic insert **22** includes a distal portion **80** and a proximal portion **82**. The proximal portion **82** can include a convex proximal surface **84** configured to interface with the patient's glenoid. The distal portion **80** includes a protrusion **86**. The protrusion **86** can be integrally formed with the proximal portion **82** such that the anatomic insert **22** is monolithic or a unitary body. The protrusion **86** is configured to interface and couple with the hole **42** of the stem face **36**. In some embodiments, the protrusion **86** couples with the hole **42** via a snap-fit, press-fit, or friction fit. In some embodiments, the anatomic insert **22** is configured to rotationally engage the stem face **36**. In the illustrated embodiment, the protrusion **86** and the hole **42** are circular and the protrusion **86** is configured to rotationally engage the hole **42**. In some embodiments, the protrusion **86** can have a tapered profile and the hole **42** can have a corresponding tapered profile such that the protrusion is configured to correspond to the hole **42**.

[0118] In some embodiments, for example as can be seen in FIGS. **6** and **15** and **17A**, the center of the hole **42** is not aligned with a center of the cavity **40** or stem face **36**, and a center of the protrusion **86** is not aligned with a center (e.g., a center of the convex proximal surface **84**) of the anatomic insert **22**. In other words, a central axis extending through the hole **42** (e.g., proximally and distally through the hole **42** or perpendicularly to the plane of the peripheral rim **38** of the stem

face **36** as shown in FIG. **13**) is offset from a central axis extending through the cavity **40** (e.g., proximally and distally through the cavity **40** or perpendicularly to the plane of the peripheral rim **38** of the stem face **36** as shown in FIG. **7**). The convex proximal surface **84** of the anatomic insert **22** can be formed about a center of rotation, and a radius R of the anatomic insert **22** can extend through the center of rotation and the center of the convex proximal surface **84**. An axis P extending through the center of the protrusion **86** can be offset from the radius R . This configuration advantageously allows for eccentric dialing of the anatomic insert **22**. In other words, this configuration allows for eccentric rotation of the anatomic insert **22** about the axis P extending through the protrusion **86** when the anatomic insert **22** is coupled to the stem **30**. Eccentric rotation moves the center of rotation of the articular surface of the anatomic insert **22** to a selected position relative to the axis extending through the protrusion. In some embodiments, the axis P of the protrusion **86** is offset from the radius R (or the center of rotation of the articular surface of the anatomic insert **22**) by five to eight millimeters. In some embodiments, the anatomic insert **22** includes a distal rim **88** (shown in FIGS. **16** and **17A**), and the protrusion **86** is offset from a center of the outer periphery of the rim **88** in a direction toward a relatively thinner region of the distal rim **88** (toward the right in FIG. **17A**). In some embodiments, part or all of the distal rim **88** of the anatomic insert **22** (shown in FIGS. **16** and **17A**) is configured to abut part or all of the peripheral rim **38** of the stem face **36** when the anatomic insert **22** is coupled to the stem face **36**. This can advantageously provide tactile feedback to the user that the anatomic insert **22** is fully coupled with the stem face **36** and provide stability to the assembled anatomic prosthesis **20**. However, in other embodiments, the distal rim **88** of the anatomic insert **22** does not abut the peripheral rim **38** of the stem face **36** in normal use to allow for adjustment of the anatomic insert **22** relative to the stem **30**. For example, when a metallic or ceramic insert is used with a metallic stem **30**, a gap is desired between the insert and peripheral rim **38** of the stem face. However, in alternate embodiments, when pyrolytic carbon is used as the insert or humeral head, contact between the metal stem and the pyrolytic carbon is preferred.

[0119] In some embodiments, the stem **30** is a fracture stem designed for use in procedures for proximal humeral fractures. An example of such a procedure is described in the Tornier Aequalis Fracture Shoulder Prosthesis Surgical Technique, which is hereby incorporated by reference herein. For example, the outer surface **35** of the proximal portion **34** could be selected based on a numerical simulation to provide accurate support to restore the tuberosities positions using virtual surgery. The stem **30** can further include a metaphyseal portion **90** between the shaft portion **32** and the proximal portion **34**, as shown in FIGS. **3-8**. The metaphyseal portion **90** includes three or more arms extending between and connecting the shaft portion **32** and the proximal portion **34**. In the illustrated embodiment, the metaphyseal portion **90** includes three arms; a medial arm **92**, a first lateral arm **94**, and a second lateral arm **96**. The medial arm **92** can be positioned near the calcar. The medial arm **92** is angled medially. In other words, a proximal end **110** of the medial arm **92** is positioned medially relative to a distal end **112** of the medial arm **92** as shown in FIG. **4B**. The first **94** and second **96** lateral arms are configured and positioned to support the tuberosities. The first **94** and second **96** lateral arms are angled outwardly or laterally. In other words, a proximal end **114** of the first lateral arm **94** is positioned laterally relative to a distal end **116** of the first lateral arm **94**, and a proximal end **118** of the second lateral arm **96** is positioned laterally relative to a distal end **120** of the second lateral arm **96** as shown in FIGS. **4B** and **5**. The angle and shape of each arms can be selected based on virtual surgery and/or a numerical simulation of bone strain due to the prosthesis to adapt to the particular patient bony anatomy. The use of the fracture stem **30** of the present disclosure in a fracture repair procedure advantageously helps promote tuberosity healing and inhibit or reduce tuberosity resorption.

[0120] The medial arm **92** can include one or more through holes **98**. The through holes **98** can be configured to receive one or more sutures in a fracture repair procedure, for example as described in the Tornier Aequalis Fracture Surgical Technique and Tornier Aequalis Reversed Fracture

Surgical Technique, each of which is hereby incorporated by reference herein. As shown in FIGS. 1-2 and 4B, the stem 30 can include a notch 100 at or near a location where a medial side 111 of the medial arm 92 meets the proximal portion 34. The notch 100 is configured to engage a suture in a fracture repair procedure and can help inhibit the suture from sliding or slipping out of position. The stem 30 can also include a fin 102 protruding from a lateral side of the shaft portion 32. In the illustrated embodiment, the fin 102 extends from a proximal portion of the shaft portion 32 distally along a portion of a length of the shaft portion 32. The fin 102 can help promote correct positioning of the stem 30 during stem placement.

[0121] FIGS. 23A-23C illustrate an embodiment of a humeral anchor 200. The humeral anchor 200 is similar to the humeral anchor 30" except as described differently below. The features of the humeral anchor 200 can be incorporated into the humeral anchor 30" and the features of the humeral anchor 30" can be incorporated into the humeral anchor 200. In some embodiments, the humeral anchor 200 can include various features described in PCT Publication No. WO 2015/112307, the entirety of which is hereby incorporated by reference herein.

[0122] The humeral anchor 200 includes a base member 204 and an anchor component 208. The base member 204 has a distal end 216 configured to be embedded in bone. The base member 204 has a proximal end 220 configured to be disposed at a bone surface. The base member 204 having a plurality of spaced apart arms 224.

[0123] The humeral anchor 200 is configured to be secured to the humeral head 22 or any another anatomic insert described herein. For example, the humeral anchor 200 can comprise a concave member 230 comprising a hole 234 configured to receive the protrusion 86 or a distal shaft of an anatomical shoulder insert. In one embodiment, the concave member 230 comprises a body coupled with the arms 224 at or adjacent to the distal end 216 of the base member 204. A continuous expanse of material can be provided between the arms 224 and the body of the concave member 230. The body and the arms 224 can be provided as a monolithic structure. The proximal end of the body of the concave member 230 can comprise an opening into which the hole 234 extends. The proximal end of the concave member 230 can include other features of the raised portion 44 of the stem 30. Other features of the proximal portion of the stem 30 that can be incorporated into the proximal end of the concave member 230 include that the proximal end of the concave member 230 can be in the same plane or substantially the same plane as the proximal end 240 of the anchor component 208, just as the raised portion 44 can be in the same plane as the peripheral rim 38 of the stem face. Also, the concave member 230 can be non-circular and radially asymmetric, e.g., having a flat peripheral edge to resist rotation or circular or non-circular but rotationally symmetric if rotation of the reverse insert is desired. Also, the sidewall of the cavity could be non-circular but rotationally symmetric in some embodiments. The disclosure of these and other features, including flats surfaces to resist rotation, should be considered as incorporated herein. The concave member 230 is enclosed at a distal end of the hole 234. The enclosed distal end of the hole 234 can include a tool interface 236, as shown in FIG. 23B. The tool interface can aid in coupling a tool for placement of the base member 204 with the proximal humerus.

[0124] The anchor component 208 has a proximal end 240 and a distal portion 244 advanceable into the base member 204 to a position disposed within the arms 224. In the embodiments depicted in FIGS. 23-23C, the distal portion 244 of the anchor component 208 comprises threads 252. The distal portion 244 can also include a cylindrical sleeve 254. The threads 252 can project laterally from the cylindrical sleeve 254. The inner periphery of the sleeve 254 can be selected to allow the anchor component 208 to be advanced over the body of the concave member 230. For example, the sleeve 254 can have a circular inner periphery that has a diameter larger than the outer circular periphery of the body forming the concave member 230. The relative sizes of the outer periphery of the body forming the concave member 230 and the sleeve 254 can be selected such that there is no resistance to advancement of the sleeve 254 relative to the body but such that a close fit is provided, e.g., a slip fit.

[0125] The threads **252** project circumferentially into a space between the arms **224** when the anchor component **208** is disposed within the base member **204**. The threads **252** are exposed between the arms **224** when the anchor component **208** is advanced into the base member **204**. As such the threads **252** are able to project into the cancellous bone in the proximal humerus to create secure connection between the humeral anchor **200** and the bone tissue. This structure creates a secure connection even before any ingrowth of bone matter into the anchor, e.g., initially at implantation of the humeral anchor **200**. This improved initial pullout force greatly reduces the chance of dislodgement, as discussed below in connection with FIG. **23D**.

[0126] The proximal end **240** of the anchor component **208** partly defines the proximal face of the humeral anchor **200**. The proximal end **240** includes features for securing the reverse insert **12** or another reverse shoulder humeral component at the proximal face. For example, a cavity **270** can be provide in the humeral anchor **200** that is at least partially defined by the anchor components **208**. In one embodiment, the cavity **270** located at the proximal face of the humeral anchor **200** has an outer periphery defined by an outer peripheral sidewall **274**. The cavity **270** has a distal wall **278** extending radially inwardly from a distal portion of the sidewall **274**. The distal wall **278** can extend from the sidewall **274** to an aperture **282** of the anchor component **208**. The aperture **282** is configured to be advanced over the proximal end of the concave member **230**.

[0127] The outer peripheral sidewall **274** can include any suitable feature for securing an insert, e.g., a reverse insert, therein. As discussed above, a C-ring, an interference fit, a fastener or other securement device can be provided for securing an insert. FIG. **23B** shows that one or a plurality of ridges **276** can be provided on the outer peripheral sidewall **274**. The ridges **276** are sized to define an inner periphery that is equal to or preferably smaller than the outer periphery of the distal projection of the insert **12**. The size difference is that sufficient to create an interference fit in some embodiments. Preferably the ridges **276** extend entirely around the outer peripheral sidewall **274**. The ridges **276** could extend along arcs including less than 180 degrees of the inner periphery. The ridges **276** could extend along arcs including less than 90 degrees of the inner periphery. The ridges **276** could extend along arcs including less than 45 degrees of the inner periphery. The ridges **276** could extend along arcs including less than 30 degrees of the inner periphery. The ridges **276** could extend along arcs including greater than 180 degrees of the inner periphery.

[0128] FIG. **23A** shows that the proximal face of humeral anchor **200**, which is defined in part by the anchor component **208**, includes a driver interface **286** disposed thereon. The driver interface **286** can be formed on the distal wall **278** in an expanse thereof disposed between the aperture **282** and the sidewall **274**. The driver interface **286** provides a convenient connection point for a tool to secure to the anchor component to enable the anchor component to be advanced by action of the threads **252** into the bone. The driver interface **286** can include two apertures that are formed through the distal wall **278**. The apertures can be simultaneously engaged by a tool and can each bear about one-half of the torque applied to the anchor component **208** to advance it into the base member **204**.

[0129] Although the threads **252** can be self-tapping in that they cut their own path into bone, in some cases a guide structure **290** is provided on the base member **204**. The guide structure **290** can include a plurality of, e.g., two, three or more than three posts **294** that project from the base member **204** into the threads **252** and ride along the threads as the anchor component **208** is being advanced into the base member **204**. In one embodiment, the guide structure includes a post **294** disposed on a side of each of the arms **224** and projecting inwardly into a space within the arms **224**. In one embodiment, the anchor component **208** has an arcuate, e.g., a semicircular, wall **298** disposed at the ends of the threads **252**. The wall **298** is configured to receive the posts **294** when the anchor component **208** is fully advanced. That is the posts come to rest against or adjacent to the wall **298**. Further the posts **294** can be viewed through apertures **292** formed in the sidewall **274** of the anchor member **208**. When the posts **294** are disposed against the wall **298** and visible through the apertures **292**, the anchor component **208** is shown to be fully advanced within the base

member **204**.

[0130] The rim **295** provides a sufficiently rigid connection between the arms **224** to prevent or reduce deflection of the arms **224** during torqueing of the anchor member **208**. The rim **295** positions the posts **296** to be in the proper position to be received in the apertures **292**.

[0131] FIG. **23C** shows the humeral anchor being disposed in the proximal humerus **h**. The proximal humerus **H** is resected to create an exposed face **F**. The face can be prepared by creating a countersunk region **CS** and by forming a created recess **CR**. The created recess **CR** can include radial projections that mirror the shape of the arms **224** such that the base member **204** can be urged into the cancellous bone of the proximal humerus **H** under relatively low force. In the illustrated embodiment, the base member **204** has a tool interface **304**, for example a concave recess formed along the outer periphery of the rim **295**. FIG. **23C** shows that the rim **295** can have three such concave recesses, for example one corresponding to the position of each of the arms **242**. Once so placed, the anchor component **208** can be rotated into the base member **204** as indicated by arrow **A** and as discussed above. In particular, the posts **294** can be aligned with the threads **254**. The anchor component **208** can be rotated with the posts **294** sliding in the threads **254**. Leading faces of the threads can have cutting edges **302** to clear a path for the anchor component **208** with reduced to minimal force. The anchor component **208** can be advanced until the posts **294** come to rest against the wall **298** and/or are visible through the apertures **292**.

[0132] FIG. **23D** illustrates the performance of certain embodiments of humeral anchors similar to the humeral anchor **200** compared to a prior art design. In particular, FIG. **23D** illustrates the initial pull-out force **1510** for Embodiment A, a variant of the humeral anchor **200** in which the anchor member **208** has a single continuous thread. Portions of a helical structure of the anchor member **208** project into the open area defined between the arms **224** and engage the bone thereby increasing the initial pull-out force of the humeral anchor **200** when initially placed. As shown in FIG. **23D**, the peak force corresponding to the initial pull out force **1510** of Embodiment A is at least ten times greater than the peak force corresponding to the initial pull out force **1500** of the prior art design having a base member and no anchor thread.

[0133] A stem holder or inserter can be used to hold the stem and/or impact it into bone during stem placement. FIG. **24** illustrates an example embodiment of a stem holder **300** configured to be used with an inserter. The plate includes two jaws and two springs to ensure retention with the stem and inserter. FIG. **25A** illustrates another example embodiment of a stem holder **310** configured to be used with stems having a proximal face **36** as described herein, and FIG. **25B** illustrates the stem holder **310** of FIG. **25A** coupled to the stem **30**. The stem holder **310** can be monolithic or manufactured in one piece. The stem holder **310** can be made of an elastic alloy, for example, titanium or a cobalt-chromium alloy. In the illustrated embodiment, the stem holder **310** is deformable or spring like, which allows the stem holder **310** to be self-retaining on the proximal face **36** of the stems shown and described herein. The stem holder **310** couples to the stem face **36** via a snap fit. A distal side of the stem holder includes compression features **312** as shown in FIG. **25A**. The raised portion **44** of the stem face **36** is received between the compression features **312** as shown in FIG. **25B**. As also shown in FIG. **25B**, a proximal side of the stem holder includes deformable features **314** (e.g., slots extending across the stem holder **310**) that allow the stem holder to flex to be coupled to and removed from the stem face **36**. An inserter, for example as shown and described at least in FIGS. **24A-24D** and corresponding description of WO 2014/067961, can be coupled to the stem holder **310** to insert the stem **30** into the humerus.

[0134] A fenestration or window **104** is defined between a lateral edge **113** of the medial arm **92** and medial edges **115**, **119** of the first **94** and second **96** lateral arms, respectively. In the illustrated embodiment, the hole **42** extends and is open to the window **104** as shown in FIG. **5**. In some embodiments of a fracture repair procedure, a bone graft can be placed in the fenestration **104** to help promote bone-to-stem **30** fixation. A space or gap **106**, shown in FIGS. **3** and **4C**, is defined or formed between an inner edge **117** of the first lateral arm **94** and an inner edge **121** of the second

lateral arm **96**. In the illustrated embodiment, the gap **106** has an elongated, rounded rectangular shape, although other shapes or configurations are also possible. As shown, the gap **106** can extend from the proximal portion of the shaft portion **32** to the proximal portion **34**. The space **106** may enable increased bone growth and fixation, for example enabling bone graft materials to be placed within the fenestration **104**, within the space **106**, and/or on a lateral side of the lateral arms **94**, **96**. For example, in some procedures, the stem **30** can be used in a tuberosity fixation procedure using a horseshoe graft. An example of such a procedure is described in Levy, Jonathan C. and Badman, Brian, Reverse Shoulder Prosthesis for Acute Four-Part Fracture: Tuberosity Fixation Using a Horseshoe Graft, J Orthop Trauma, Volume 25, Number 5, May 2011, which is hereby incorporated by reference herein. In such a procedure, the horseshoe graft can be placed on the lateral surface of the metaphyseal portion **90**. The design of the metaphyseal portion **90** helps provide stability to the horseshoe graft. The space **106** allows the horseshoe graft to form or grow into or through the window **106** and improve fixation and tuberosity repair. In particular, resorption of the tuberosities can be reduced or prevented because more pathways for formation or growth of bone are provided to bridge between the fractured portions. The peripheral rim **38** of the stem face **36** can also help support and stabilize the tuberosities. The shape and size of the horseshoe graft can be selected based on a numerical simulation to accurately restore the tuberosities positions using virtual surgery.

[0135] FIGS. **26A-E** illustrate a bone cutter **330** that can be used to prepare the bone graft according to embodiments of the present disclosure. To prepare the bone graft, the resected humeral head or other bone graft material **400** is placed in a base **332** of the bone cutter **330** (with the cut side facing upwards or the top when using the humeral head), as shown in FIG. **26A**. A drilling guide **334** including holes **336** is placed onto the bone graft **400**, and a drill bit **338** is used to drill holes into the bone graft **400** through the holes **336** as shown in FIG. **26B**. In the illustrated embodiment, the drilling guide **334** includes four holes **336** so that the drill bit **338** can be used to drill four holes into the bone graft **400**. The holes **336** can be provided at different angles to create two angled grooves in the bone graft **400** that allow the prepared bone graft to fit the metaphyseal portion **90** of the fracture stem **30**. Each grooves fit over the lateral arms **94** and **96** so that the inner portion of the graft fill the gap **106** and the window **104**. A cutting cap **340** is placed on top of the base **332** applying pressure to secure the bone graft **400** inside the bone cutter **300** as shown in FIG. **26C**. The cutting cap **340** is impacted on top with a mallet until the cutting cap **340** contacts the base **332** as shown in FIG. **26D** to shape a diameter of the bone graft **400** into a cylinder. A saw blade is placed through a slot **342** to further shape the bone graft into a cylinder (for example, by cutting off the rounded bottom portion of the humeral head). The bone graft is extracted by pushing with an instrument **344** as shown in FIG. **26E**. FIG. **27A** illustrates the humeral head or other bone graft material **400** prior to be prepared using the bone cutter **330**, FIG. **27B** illustrates the bone graft material **400** including angled grooves **402** after the holes are drilled using the drilling guide **334**, and FIG. **27C** illustrates the prepared bone graft **400**. The bone graft material **400** may be snap or press-fit onto the fracture stem. FIGS. **28A-28B** illustrate the prepared bone graft **400** disposed on the fracture stem **30**. A kit for a shoulder prosthesis can include a fracture stem as shown and described herein, the bone cutter **300**, a drill, a mallet, and/or bone graft material. The bone graft material can be, for example, autograft or allograft, and can be made of bone, stem cells, a ceramic, or a polymer. Additional details regarding devices and methods involving placement of bone graft during a shoulder prosthesis procedure can be found in U.S. 2008/0183297, the entirety of which is hereby incorporated by reference herein.

[0136] FIG. **18** illustrates a schematic of a natural human humerus **H**, and FIG. **19** illustrates the stem **30** implanted in the humerus **H**. As shown in FIG. **19**, the natural humeral head has been resected. Resection plane **120** is the surface from which bone has been removed and the external surface of the remaining bone. With a standard shoulder prosthesis, when a surgeon desires to convert a primary anatomic shoulder prosthesis into a reverse shoulder prosthesis, he or she must

typically remove the entire prosthesis, thereby risking further weakening the bone. However, using a modular system such as implants **10** and **20**, the surgeon may leave the stem **30** implanted in the bone and simply replace the anatomic insert **22** with the reverse insert **12**. FIGS. **20-22** illustrate an example procedure for converting a modular anatomic shoulder implant **20** into a modular reverse shoulder implant **10**. As illustrated in FIG. **21**, the anatomic insert **22** may be removed using a wedge or similar instrument **200**. The reverse insert **12** can be inserted into direct engagement with the stem face **36**, for example, impacted into place using an impactor **202** as shown in FIG. **22**. [0137] As shown in FIG. **19**, the resection plane **120** can be flush or substantially flush with the stem face **36**, i.e., with the peripheral rim **38** of the stem face **36**, when the stem **30** is implanted in the humerus **H**. The interface between the stem **30** and the anatomic insert **22** in the anatomic prosthesis and the interface between the stem **30** and the reverse insert **12** in the reverse prosthesis are substantially flush with the resection plane **120**. The modularity of the system can therefore be located at the resection plane and the conversion from an anatomic prosthesis to a reverse prosthesis can occur at the resection plane. In embodiments, instrumentation can be used to measure the angle of the resection plane so that the proper stem may be selected. The modularity of the system could alternatively be located slightly above the resection plane **120**. The surgeon may then remove the anatomic insert **22** and implant the reverse insert **12** without having to remove a component that may have integrated into the patient's bony anatomy and without having to perform additional reaming or other bone preparation. This can advantageously help reduce or inhibit excessive patient bone loss.

[0138] In an example method for shoulder surgery, a unitary stem is provided. The unitary stem includes a distal portion and a proximal portion. The proximal portion includes a stem face that is configured to directly couple to a reverse shoulder insert and an anatomical shoulder insert. A surgeon or other user chooses intra-operatively to implant the reverse shoulder insert or the anatomical shoulder insert. The surgeon or other user implants a two-component shoulder system that is selected from the group consisting of a stem directly attached to the reverse shoulder insert and a stem directly attached to the anatomical shoulder insert.

[0139] In another example method for shoulder surgery, a stem of a humeral component shoulder system is implanted at least partially in a medullary canal of a humerus. The stem is adapted to directly interface with a one-component reverse shoulder insert and an anatomical shoulder insert. A surgeon or other user chooses intra-operatively to implant a one-component reverse shoulder insert or an anatomical shoulder insert. The surgeon or other user directly couples the chosen insert to the stem.

[0140] In another example method for shoulder surgery, a unitary stem is disposed at least partially in a medullary canal of a humerus. A surgeon or other user selects a reverse shoulder insert or an anatomical shoulder insert. The reverse shoulder insert includes a body with a concave articular surface on one side and an engagement structure projecting from a side of the body opposite the concave surface. The surgeon or other user implants the insert that has been chosen by directly coupling the chosen insert to the stem.

[0141] In an example method for revision shoulder surgery, an anatomical shoulder insert is removed from a unitary stem implanted in a patient's humerus to expose a proximal face of the stem. A reverse shoulder insert is directly coupled to the proximal face.

[0142] Implants of the present disclosure may be designed and/or manufactured with computer-assisted image method. Patient specific implants (PSI) include, briefly, obtaining images of patient anatomy through, for example, MRI, CT, or x-ray images. These images may then be used to create patient specific prosthesis components including stems, humeral heads, reverse inserts, and stemless shoulder components/implants. Further, patient specific guides, instruments such as graft cutter, and templates can also be provided by various CAD programs and/or commercially available software. The PSI are generally formed using computer modeling for matching a three-dimensional image of the patient's anatomy by the methods discussed above.

[0143] Alternatively PSI imaging may be used to provide a virtual model of the patient's anatomy, enabling surgeons to see in three-dimensions how a specific implant will fit into the patient's anatomy. The implant and/or the associated graft can then be altered or tailored to the surgeon's preference prior to sending along the imaging data to manufacture the implant, graft, guides, instruments, or template.

[0144] With the modular system of the present disclosure, the surgeon can advantageously decide intra-operatively whether to implant the anatomic insert **22** or reverse insert **12**. In some cases, a surgeon may believe that a patient is a candidate for an anatomic prosthesis based on, for example, pre-operative examination and imaging, and may prepare for such a procedure. However, during the procedure, the surgeon may determine that a reverse prosthesis is necessary or desired instead. With some previously available prostheses, in such a situation the surgeon may need to obtain an entirely different prosthesis, which may not be possible once the surgery has begun. Even if the surgeon pre-operatively selected a modular system, the surgeon may need to prepare the bone differently for the reverse prosthesis component(s) and may not have the appropriate tools, resources, or time to do so if an anatomic prosthesis procedure was planned. With the systems of the present disclosure, the surgeon can select which insert to use, or can change from a planned anatomic prosthesis to a reverse prosthesis, during the procedure. Further, the kit of the present disclosure may be used when converting from an anatomical shoulder system to a reverse shoulder system.

[0145] In some embodiments, a kit includes the stem **30**, the anatomic insert **22**, and the reverse insert **12**. Providing both inserts **12**, **22** in a kit can advantageously allow the surgeon to select whether to use the anatomic insert **22** or reverse insert **12** pre-operatively or intra-operatively.

[0146] Some embodiments have been described in connection with the accompanying drawings. However, it should be understood that the figures are not drawn to scale. Distances, angles, etc. are merely illustrative and do not necessarily bear an exact relationship to actual dimensions and layout of the devices illustrated. Components can be added, removed, and/or rearranged. Further, the disclosure herein of any particular feature, aspect, method, property, characteristic, quality, attribute, element, or the like in connection with various embodiments can be used in all other embodiments set forth herein. Additionally, it will be recognized that any methods described herein may be practiced using any device suitable for performing the recited steps.

[0147] For purposes of this disclosure, certain aspects, advantages, and novel features are described herein. It is to be understood that not necessarily all such advantages may be achieved in accordance with any particular embodiment. Thus, for example, those skilled in the art will recognize that the disclosure may be embodied or carried out in a manner that achieves one advantage or a group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

[0148] Although these inventions have been disclosed in the context of certain preferred embodiments and examples, it will be understood by those skilled in the art that the present inventions extend beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the inventions and obvious modifications and equivalents thereof. In addition, while several variations of the inventions have been shown and described in detail, other modifications, which are within the scope of these inventions, will be readily apparent to those of skill in the art based upon this disclosure. It is also contemplated that various combination or sub-combinations of the specific features and aspects of the embodiments may be made and still fall within the scope of the inventions. It should be understood that various features and aspects of the disclosed embodiments can be combined with or substituted for one another in order to form varying modes of the disclosed inventions. Further, the actions of the disclosed processes and methods may be modified in any manner, including by reordering actions and/or inserting additional actions and/or deleting actions. Thus, it is intended that the scope of at least some of the present inventions herein disclosed should not be limited by the particular disclosed embodiments described above. The

limitations are to be interpreted broadly based on the language employed and not limited to the examples described in the present specification or during the prosecution of the application, which examples are to be construed as non-exclusive.

Claims

1. A humeral implant system, comprising: a stem including: a proximal portion having a hemispherical shape, the proximal portion defining an interior cavity circumscribed by a proximal rim, wherein the interior cavity is configured to receive either of a reverse insert having a concave proximal surface and an anatomic insert having a convex proximal surface; and a second portion extending from the proximal portion of the stem along a longitudinal axis, the second portion sized and configured to be received in at least a metaphyseal portion of a humerus, the second portion including at least three arms extending radially away from the longitudinal axis, wherein the second portion of the stem tapers along its length such that a distal tip of the stem is narrower than a proximal end of the second portion of the stem.
2. The humeral implant system of claim 1, wherein the proximal portion defines a recess within the cavity.
3. The humeral implant system of claim 2, wherein the recess extends through the proximal portion of the stem.
4. The humeral implant system of claim 3, wherein a window is defined between the arms and the recess is open to the window.
5. The humeral implant system of claim 1, wherein at least one of the arms defines a hole configured to receive one or more sutures.
6. The humeral implant system of claim 1, further comprising: the reverse insert, the reverse insert including a protrusion sized and configured to be received in the internal cavity defined by the proximal portion of the stem.
7. The humeral implant system of claim 6, wherein an inner periphery of the proximal portion that defines the internal cavity includes a groove for securing the reverse insert to the stem.
8. The humeral implant system of claim 1, further comprising: the anatomic insert, the anatomic insert including a surface configured to abut the peripheral rim of the stem.
9. The humeral implant system of claim 1, wherein the internal cavity defined by the proximal portion of the stem extends from the peripheral rim to a planar base surface.
10. A humeral implant system, comprising: a reverse insert having a concave proximal surface and a protrusion extending distally; and a humeral anchor sized and configured to be received in at least a metaphysis of a humerus, the humeral anchor including: a first portion having a hemispherical shape and defining a cavity extending inwardly from a proximal rim and terminating a planar base, the cavity sized and configured to receive the protrusion of the reverse insert therein, the first portion of the humeral anchor further adapted to be coupled to an anatomic articulation component having a convex surface; and a second portion extending from an external surface of the hemispherical shape of the first portion and defining a longitudinal axis, the second portion including at least three members each of which extends from the external surface of the first portion toward a distal end of the second portion, wherein the second portion of the humeral anchor tapers along its length such that the distal end of the second portion is narrower than a proximal end of the second portion.
11. The humeral implant system of claim 10, wherein an inner periphery of the first portion of the humeral anchor that defines the cavity includes a groove for securing the reverse insert to the stem.
12. The humeral implant system of claim 11, wherein the first portion of the humeral anchor defines a recess within the cavity for securing the anatomic articulation component to the humeral anchor.
13. The humeral implant system of claim 12, wherein the recess extends through the hemispherical

shape of the first portion of the humeral anchor.

14. The humeral implant system of claim 12, further comprising: the anatomic articulation component, the anatomic articulation component including a surface configured to abut the peripheral rim of the humeral anchor.

15. The humeral implant system of claim 10, wherein at least one member of the at least three members of the humeral anchor defines a hole sized and configured to receive one or more sutures.

16. The humeral implant system of claim 10, wherein the at least three members includes: a first member arranged medially with respect to the longitudinal axis; and two members arranged laterally with respect to the longitudinal axis and the first member.

17. The humeral implant system of claim 16, wherein the first member is angled medially such that a proximal end of the first member is positioned medially relative to a distal end of the first member.

18. The humeral implant system of claim 17, wherein the two members are angled outwardly such that a proximal end of each of the two members is positioned laterally relative to a distal end of each of the two members.

19. An implant system, comprising: a stem component sized and configured to be disposed in a humerus, the stem component including a proximal end having a hemispherical shape and defining a cavity that extends inwardly from a proximal face, a plurality of members from an external surface of the proximal end in a distal direction, the plurality of members including: a first member arranged medially with respect to a longitudinal axis defined by a distal portion of the stem; and two members arranged laterally with respect to the longitudinal axis and the first member, wherein the cavity includes engagement features for coupling a reverse articular component and an anatomic articular component to the stem.

20. The implant system of claim 19, wherein the engagement features includes: a first engagement feature for coupling the reverse articular component to the stem, the first engagement feature including a circumferential groove defined by an interior surface of the proximal end of the stem that defines the cavity; and a second engagement feature for coupling the anatomic articular component to the stem, the second engagement feature including a recess defined by the proximal end of the stem within the cavity.
